PHASE 7 DOCUMENT

EduGenie: Google Gemini Powered Learning Assistant

Phase Name: System Testing, Quality Assurance and Security Validation

Project Name	EduGenie: Google Gemini Powered Learning Assistant
Phase	Phase 7 – Testing, QA and Security Validation
Project Type	AI-Powered Learning Application
Main Technology	Google Gemini API
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 7 of **EduGenie: Google Gemini Powered Learning Assistant**. This phase verifies that the modules developed in earlier phases work correctly together. It covers functional testing, integration testing, usability checks, performance observations, security validation, defect handling, and user acceptance preparation. The purpose is to identify and correct issues before final deployment and project evaluation.

2. Phase Objective

The objective of this phase is to confirm that EduGenie meets the defined requirements and provides a reliable learning experience. Testing verifies authentication, user roles, AI learning chat, quiz generation, study planning, progress tracking, project assistance, faculty resources, and secure Gemini integration. Quality assurance ensures that the system is understandable, stable, responsive, and safe for its intended educational use.

3. Testing Scope

The testing scope includes student registration and login, role-based access, dashboard navigation, Ask EduGenie requests, learning-mode selection, Gemini response display, saved content, quizzes, quiz attempts and feedback, study plans, progress data, project guidance, faculty resource publishing, student resource access, backend error handling, API-key protection, and mobile-responsive behavior. External services, such as Gemini availability, are tested through controlled request and error scenarios.

4. Test Environment

Testing can be performed in a development or staging environment that is separate from production data. The environment includes the EduGenie frontend, backend API service, test database, configured Gemini API credentials, supported browsers, and representative mobile devices or emulators. Test accounts should be created for student, faculty, and administrator roles. Test data should not contain unnecessary real personal information.

5. Functional Testing

Functional testing confirms that each feature performs its intended action. Students should be able to log in, ask learning questions, select learning modes, view answers, save content, generate quizzes, submit quiz attempts, create study plans, and request project guidance. Faculty users should be able to publish and manage authorized resources. Administrators should be able to manage roles and settings as permitted. Each function is tested with valid, invalid, and boundary inputs.

6. Integration Testing

Integration testing verifies that system modules work together. Important flows include login to dashboard navigation, frontend request to backend validation, backend communication with Google Gemini, Gemini response to frontend display, quiz generation to attempt storage, study-plan generation to task tracking, faculty resource publication to student access, and project-guidance output to saved student drafts. Failed external AI responses should not break the application interface.

7. Sample Test Cases

1. Valid student login: Expected result – the student dashboard opens successfully.
2. Invalid login: Expected result – access is denied with a clear error message.
3. Ask EduGenie with a valid question: Expected result – the system returns a formatted learning response.
4. Empty AI question: Expected result – the application requests a valid question without calling the AI service.
5. Quiz generation: Expected result – topic-based practice questions are displayed.
6. Quiz submission: Expected result – answers and feedback are recorded for the correct student.
7. Study-plan generation: Expected result – a plan is displayed from valid goals and available time.
8. Faculty resource publication: Expected result – authorized students can view the published resource.
9. Unauthorized resource edit: Expected result – access is denied.
10. Gemini service failure: Expected result – the user receives a friendly retry message without system failure.

8. Usability and Accessibility Testing

Usability testing checks whether students can understand navigation, labels, buttons, messages, and learning outputs without unnecessary guidance. Test participants should be able to find the chat, quiz, study planner, saved content, and project-assistance features. Accessibility checks include readable text, sufficient visual contrast, responsive layout, keyboard navigation where practical, meaningful form labels, error messages that explain what to do next, and mobile-friendly touch controls.

9. Security Validation

Security validation confirms that only authorized users can access protected features and data. Checks include password and session protection, role-based permissions, HTTPS usage, server-side Gemini API-key storage, input validation, output encoding, protection against unauthorized resource access, safe error messages, restricted administrative access, and removal of sensitive configuration files from public repositories. The system should test that a student cannot view another student's saved learning data or project drafts without permission.

10. Performance and Reliability Checks

Performance checks observe response time for common pages and AI requests under normal usage. The system should show a loading state while waiting for Gemini and should handle slow or failed requests gracefully. Reliability testing includes repeated requests, session expiry, network interruption, database errors, and recovery after service availability returns. Rate controls can be checked to prevent excessive automated requests from affecting normal student access.

11. Defect Management

Any issue found during testing should be documented with a unique identifier, module name, test condition, expected result, actual result, severity, status, assigned person, and resolution notes. High-severity defects involving security, incorrect access control, data exposure, or core feature failure should be resolved before release. After a fix is applied, the related test should be executed again to confirm that the defect is resolved and no regression has been introduced.

12. User Acceptance Testing

User acceptance testing involves selected students and faculty members using EduGenie in realistic learning scenarios. They provide feedback on whether features are useful, understandable, and aligned with academic needs. Example activities include asking a subject doubt, attempting a quiz, creating a study plan, requesting project guidance, and accessing a faculty-shared resource. Feedback is reviewed to make final improvements before deployment.

13. Test Results and Acceptance Criteria

The phase is considered successful when all critical user flows pass, high-severity issues are resolved, user-role restrictions operate correctly, the Gemini API key remains protected, error handling is understandable, student data remains private, and selected users can complete the main learning tasks. Minor visual improvements may be documented for future releases if they do not affect functionality or security.

14. Phase Deliverables

The deliverables are the test plan, test cases, test-data setup, functional test results, integration-test results, usability and accessibility observations, security-validation checklist, performance observations, defect log, retest records, user-acceptance feedback, and a release-readiness summary.

15. Planned Next Phase

The next phase will be **Phase 8: Deployment, Final Documentation and Project Evaluation**. It will prepare the production deployment, user guidance, final project report, demonstration material, evaluation checklist, and future-enhancement plan for EduGenie.

16. Conclusion

Phase 7 ensures that EduGenie is tested as a complete AI-powered learning application rather than as separate modules. Through functional, integration, usability, performance, and security testing, the project can deliver a dependable and responsible platform for students and faculty. The validated system is then ready for final deployment and evaluation.

Phase Approval

Prepared by	_____
Guide / Faculty	_____
Approved by	_____